

Summing using De Moivre's

Key Points

De Moivre's theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

Using exponential form often makes it much easier when summing

$$z = e^{i\theta}$$

If your sum contains sine use the imaginary part, if your sum contains cosine use the real part

Basic Skills

Q1. a) State De Moivre's theorem

b) Expand $(\cos \theta + i \sin \theta)^4$

c) State $\operatorname{Re}((\cos \theta + i \sin \theta)^4)$

d) Hence write an expression for $\cos 4\theta$

e) State $\operatorname{Im}((\cos \theta + i \sin \theta)^4)$

f) Hence write an expression for $\sin 4\theta$

Q2. Use De Moivre's theorem to find an alternate expression for

a) $\cos 6\theta$

b) $\sin 12\theta$

Competency Skills

Q3. Given that $C = 1 + \cos \theta + \cos 2\theta + \dots + \cos 10\theta$ and $S = \sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin 10\theta$

a) Show that $C + iS = 1 + z + z^2 + \dots + z^{10}$ where $z = \cos \theta + i \sin \theta$

b) Show that $C + iS = \frac{1 - z^{11}}{1 - z}$ where $z = \cos \theta + i \sin \theta$

c) By rationalizing show that $C + iS = \frac{1}{2(1 - \cos \theta)} (1 - \cos \theta + \cos 10\theta - \cos 11\theta + i(\sin \theta + \sin 10\theta - \sin 11\theta))$

d) Hence deduce C and S

e) Hence find $C = 1 + \cos \theta + \cos 2\theta + \dots$ and $S = \sin \theta + \sin 2\theta + \sin 3\theta + \dots$, the infinite versions

Q4. Find the series' $C = 1 + \frac{1}{2}\cos \theta + \frac{1}{4}\cos 2\theta + \dots$ and $S = \frac{1}{2}\sin \theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \dots$

Q5. Show that $\cos \theta + \cos\left(\theta + \frac{2\pi}{n}\right) + \cos\left(\theta + \frac{4\pi}{n}\right) + \dots + \cos\left(\theta + \frac{2(n-1)\pi}{n}\right) = 0$ for any n

Advanced Skills

Q6. a) By using double angle formula show that $1 + e^{i\theta} = 2e^{\frac{i\theta}{2}} \cos \frac{\theta}{2}$

b) For $C = 1 + \binom{n}{1}\cos \theta + \binom{n}{2}\cos 2\theta + \dots + \binom{n}{n}\cos n\theta$ create the corresponding series for sine, S

c) Show that $C + iS = (1 + e^{i\theta})^n$

d) Hence find C and S

Q7. Use De Moivre's theorem to find

a) The series $S = 1 + 2\sin x + \frac{2^2 \sin 2x}{2!} + \frac{2^3 \sin 3x}{3!} + \dots$

b) What other series can you deduce using part a)

Extension

Q8. Explain why we need de Moivre's theorem to work out sine and cosine series

Hint: Look over your answers to see where it's important

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